DATA COLLECTION SYSTEMS

CANADIAN LIDAR

AlignPro

Lidar Boresight Calibration

Stage 02 of the DCS chain: PosPro → AlignPro → CloudPro

User Guide

for AlignPro 1.1.3

AlignPro — User Guide

Canadian Lidar · AlignPro 1.1.3 · November 2026

About AlignPro

AlignPro finds the Lidar mounting angles (roll, pitch, heading) of the Canadian Lidar mobile mapper from a short calibration drive, and hands them to CloudPro and to the mapper. This guide is for **AlignPro 1.1.3**.

It is stage 02 of the Data Collection Systems chain. Each stage reads only what the stage before it published:

Stage	Plugin	Reads	Publishes
01	PosPro	mapper captures (.anpp, .clraw)	trajectory.json + SBET
02	AlignPro	captures (.clraw) + PosPro's trajectory	alignment.json
03	CloudPro	captures + trajectory + alignment	cloud.json + LAS

AlignPro reads the `.clraw` captures directly. It splits them into passes, finds flat surfaces (walls, roofs, ground) that several passes see, and solves the three angles so those surfaces line up. It solves again with each line left out in turn; the spread of those answers is the SDEV. A live Data View shows plan, profile and section of the points while you adjust the values by hand.

What's new

Version	Date	Change
1.1.3	2026-11-03	This user guide in the plugin: **?** button, project menu, Help › Plugins
1.1.2	2026-11-02	Panels dock only from the title bar's right-click menu; dragging just moves them
1.1.1	2026-11-02	QGIS 4 no longer closes with an error after a panel was floated as a window
1.1.0	2026-11-02	Part of the DCS chain: shared job project, PosPro's trajectory, publishes alignment.json

Version	Date	Change
1.0.0	2026-11-02	First release: auto calibration, Data View, quality control, reports

Install

Install **DCS Core** first, then AlignPro, then restart QGIS once. AlignPro needs QGIS 3.34 or later; QGIS 4 works too.

1. In QGIS: **Plugins › Manage and Install Plugins › Install from ZIP**, choose `dcs_core-<version>.zip`, Install.
2. Same again with `alignpro-1.1.3.zip`. Replace an older AlignPro if asked.
3. Restart QGIS.
4. Open AlignPro from **Plugins › Canadian Lidar › Data Collection Systems › AlignPro**, or its toolbar button.

No other Python packages are needed. Without DCS Core, the AlignPro button only shows a message asking for it.

This guide inside QGIS. The plugin carries this guide as a PDF for its own version. Open it from:

- the **?** button at the top right of the AlignPro panel, beside the window button;
- the project box menu: **User guide (this version)**;
- **Help › Plugins › AlignPro user guide**.

Before you start: the calibration drive

A good result needs six passes past solid vertical surfaces, driven in opposite directions at different distances. The crew records it like any other job.

- **Site:** building walls, a building corner or thick poles beside open, flat ground. Walls facing the road and walls across it are both useful: across-road walls fix heading.
- **Passes:** at least 3, ideally 6, as reciprocal pairs (out and back) at increasing distance from the target. Drive steadily; 20 to 40 km/h is fine.
- **One pass per Start/Stop** is easiest. Several passes in one capture also work: AlignPro splits them at the turns and drops the stops.

- **File split:** set it large enough that a pass is never split across files (3.9 GB is about 40 minutes of .clraw).
- **Positioning:** good sky view, a base station within 20 km, and 5 minutes standing still at the start and the end, so PosPro can make a fixed trajectory.
- **Before driving:** enter the measured Lidar lever arm in the mapper (Settings › Hardware › Survey sensor). AlignPro solves the angles, not the lever arm.
- **During the drive:** check that the Lidar time lock (PPS) stays green, and press Mark when passing the target.

Process the job in PosPro first. AlignPro can start on the mapper's real-time trajectory for a quick look, but the final calibration must use PosPro's.

Projects

AlignPro works on the job's shared processing project, the same one PosPro and CloudPro use. Whichever plugin is used first creates it; the others open it.

```
JobFolder/                                  mapper captures, never changed
  Processing/<Name>/<Name>.dcsproj          the shared project
    01_PosPro/out/trajectory.json           PosPro's trajectory
    02_AlignPro/alignpro.json               AlignPro's settings and results
                work/                       caches, may be deleted
                out/                        alignment.json, reports
                log/                        one log per run
    03_CloudPro/
```

- **Open project…** (project box, top right): choose the `.dcsproj` PosPro made. **Recent projects** lists the last ones; AlignPro reopens the last project when QGIS starts.
- **New project…**: choose the job folder that holds the captures and a name. AlignPro offers the `.clraw` files it finds there.
- **Captures:** AlignPro uses the captures recorded in the project and finds their `.clraw` files. Extra files can be added on the Data page.
- **Raw data moved?** Project box › **Relocate the job folder**. Nothing goes out of date because of the move.
- **Save the QGIS project in the project folder** keeps the map layers with the job.
- Older stand-alone AlignPro projects (`.alignpro.json`) still open, without the chain bar.

The AlignPro window

Work down the seven steps on the left; each step's pill turns green when done and amber when something needs a look.

- **Header:** the project box (project menu), the **?** button (this guide) and the window button ⤢ (maximise, float as a window, dock right or left, open the Data View).
- **Chain bar** under the header: 01 PosPro → 02 AlignPro → 03 CloudPro, each marked done (✓), out of date (⚠ stale), running, failed or not run. Hover a stage for the reason; click another stage to open that plugin on this project.
- **Steps:** 1 Data, 2 Installation, 3 Quality, 4 Pairs, 5 Calibration, 6 Analyze, 7 Report. The pill's second line says what each holds.
- **Under the steps:** the **Data View · Fine Tune** button, the live average error with the current angles, and **Map grid** (which grid the map shows).
- **Footer:** short messages, the status line, a progress bar and **STOP** while a job runs.

Panels and docking. The AlignPro panel and the Data View dock only when you ask. Right-click a panel's title bar and choose **Dock right**, **Dock left**, **Dock bottom** or **Dock top**, or **Float**, **Float as a window**, **Maximise**, **Close**. Dragging a panel just moves it. Double-click the title of a docked panel to float it.

On the map: AlignPro adds a group with the pass tracks (one colour per line, the same colours everywhere), line starts, the calibration boxes, the flat surfaces it uses and a grid (line separation, standard deviation, mean height with hillshade, or point count).

1 Data

The Data page shows which captures and which trajectory the calibration uses, and lists the passes. The first read decodes every pass; later opens read from the cache in `work/`.

- **Job folder and captures:** taken from the shared project. Missing files are flagged with a pointer to Relocate. **Add files… / Add folder…** add more `.clraw`; **Read files** reads everything again.
- **Trajectory:**
  - **PosPro trajectory published for this project** (the default). The line under it says whether it is current or out of date.
  - **Another post-processed SBET file**, with its date and leap seconds.
  - **Certus real-time**, for a quick look only: without RTK fixed throughout, the position wanders between passes and the result absorbs it.

- **What the files contain:** recording times, passes, Lidar model and return mode, channel calibration, points kept per line, trajectory and mapper version.
- **Lines table:** one row per pass with direction (COG), speed, length, RTK-fixed %, PPS lock %, lost packets, points, median range and the error before and after calibration. Untick **Use** to leave a pass out; pairs and surfaces are found again.

A pass needs at least 4 seconds of straight driving. A capture with turns and stops is split into passes automatically (names like `003a`, `003b`).

2 Installation

The Installation page holds the Lidar lever arm, latency and mounting angles the calibration starts from. They come from the capture's hardware record; the drawing shows the vehicle with the Certus, both antennas and the Lidar.

- **Lever arm (m):** Certus reference point to the Lidar optical centre, Forward, Starboard, Down. It is measured, not solved. A lever arm of 0, 0, 0 is flagged: enter the measured one.
- **Mounting angles (deg):** roll, pitch, heading in the mapper's convention (see Conventions). The values recorded in the capture are shown under each.
- **Lidar latency (ms):** leave at 0 while the PPS time lock is good.
- **Apply** re-georeferences every pass with these values and publishes them (see What AlignPro publishes). **Revert to recorded** restores the capture's values.
- **Certus as recorded:** model, alignment and antenna offsets, for reference.

A corrected lever arm is kept in AlignPro's results only. To make it permanent, enter it in the mapper.

3 Quality

The Quality page checks the data before you trust a calibration. The four cards at the top give the answer: PPS lock, lost Lidar packets, RTK fixed and the mean error now.

Section	What to look for
Time sync and rates	PPS lock 100 %, about 5000 Lidar packets/s, arrival minus sensor time about 1 ms, no lost packets

Section	What to look for
Positioning	position accuracy steady and small over each pass; fix type RTK fixed
Attitude accuracy	roll, pitch and heading accuracy steady, no jumps
Heading vs course	almost no crab on a land vehicle; a steady offset in both directions points at the trajectory heading
Residuals	the current curve narrow and centred on 0, the recorded curve wider
Ranges	points spread over the range gate, not all close to the vehicle

A lost PPS lock or many lost packets on a pass is a reason to leave that pass out (Data page, **Use**).

4 Pairs

The Pairs page shows which passes overlap and which flat surfaces the calibration will use. The step turns green with at least 2 reciprocal pairs and 20 wall patches.

Pair type	Meaning
Reciprocal	opposite directions over the same ground: the strongest for roll and pitch
Same direction	same direction, side by side
Crossing	passes at an angle
Latency	same direction, clearly different speeds: shows a timing error
Additional	too little overlap or too few shared surfaces; not used

A **patch** is a 2 m piece of a flat surface that two or more passes see. Walls drive heading, so flat ground is capped to the number of wall and sloped patches. The map layer *Planar patches* shows them.

- **Draw area of interest:** for a single target (a pole, a building corner), draw around it on the map; only patches inside are used. **Clear area** goes back to automatic.
- **Place boxes automatically** puts the roll, pitch and heading inspection boxes on suitable walls. **Free profile** draws a profile anywhere.
- **Settings…** holds the data gates (range, return, points per line), pass splitting, pairing limits, patch size and solver options. The defaults suit the mapper.

5 Calibration

Auto calibrate solves roll, pitch and heading from every patch, then once more with each pass left out. On the test data it recovered a known error of 0.20, −0.30 and 0.45° to within 0.0005°, and the mean error between passes fell from 4.08 to 0.29 cm.

1. Press **Auto calibrate**. The panel shows the advised angles, each with its SDEV, and the error before → after.
2. Read the cards. SDEV under 0.02° is green, 0.02 to 0.05° amber, above red. The error after should be about 1 cm or less.
3. Check the **Data View** (it opens with the advised angles over the applied ones in grey).
4. Press **Apply advised**. The angles are applied and published for CloudPro.
5. Press **Copy for mapper** and paste into the mapper: Settings › Hardware › Survey sensor › Mounting angles.

Calibrating alone publishes nothing; only applying does.

- **Combinations table:** one row per solve (all lines, then each line left out). Select a row to try its angles in the Data View; right-click to apply them or copy them.
- **Boresight about vehicle axes:** the correction from the recorded mounting, with its SDEV. Use it near pitch 90°, where roll and heading trade against each other (see Conventions).
- **Current angles:** type values and press **Apply** to use your own.
- **Published for CloudPro:** what was last published, when, and whether it is current. **Publish the applied values** appears when the applied values differ from the published ones.

Warnings appear in amber, for example few walls (heading weakly fixed), a large SDEV, or a solve on the real-time trajectory.

Data View and fine tuning

The Data View shows the real points in plan, profile and section and redraws them about 0.1 s after every change of angle or lever arm. Use it to check the automatic result and to fine-tune by hand when it is wrong. Open it from **Data View · Fine Tune** (sidebar or Calibration page); it also opens after loading and after Auto calibrate.

- **Plan** (top left): up to 400 000 points around the target, true scale. Right-drag draws a new profile. **Plan height band** hides the ground so walls and poles stand out.
- **Profile** (top right): along the chosen box. For the Heading box it shows the wall from above, edge-on: a heading error turns the passes apart.
- **Section** (bottom right): across the profile, at the point you click in the profile.
- **Section** list: Roll box, Pitch box, Heading box, or Custom. **Profile width** and **Section** thickness set how much is shown.
- **Ghost:** the same points in grey with the recorded, applied or advised values, or as the view opened. Coloured points are the trial values; grey shows where they were.
- **Colour:** by line (the line colours), height or intensity.
- Wheel zooms, Shift + wheel stretches the height, drag pans, double-click fits. **Load view area** loads the points around the plan's centre; **Fit** fits all three views.

Adjusting by hand. Each value has − and + buttons (hold to repeat) and a **Step** (0.001 to 1° for angles, 0.001 to 0.1 m for the lever arm); you can also type a value. Latency changes take about half a second.

1. **Start from** Recorded, Applied or Advised.
2. Pick the box that shows the angle you want: roll in the Roll box, pitch in the Pitch box, heading in the Heading box.
3. Nudge until the passes fall on one line. The **Error now** card follows each step and shows the change against the applied values.
4. **Apply to project** publishes the values; **Copy for mapper** copies the three angles.

Until you apply, the panel says *Trial values not applied yet*.

6 Analyze

Analyze sweeps one value with everything else fixed and plots the mean error against it; the lowest point is the best value. Use it to check latency or a lever-arm component, or to confirm an angle.

1. Choose the **Parameter**: Lidar latency, lever forward, starboard or down, or one of the three angles.
2. Set **Centre**, **Range ±** and **Steps**, then press **Analyze**.
3. **Apply best value** applies and publishes it.

If the curve is jumpy, widen the range and add steps. Small values found on a quiet drive are usually noise: leave a working setup alone.

7 Report

The Report page writes the calibration report as PDF and text into `02_AlignPro/out`, named `Report_YYYY-MM-DD-HHMMSS_<project>`.

- **Header:** vehicle, operator, client, location and notes.
- **Options:** add the header, ask for a file name, add screenshots (the map, the Data View and the three boxes), copy the text to the clipboard.
- The report holds the recorded, starting and advised angles with SDEV, the boresight correction, the error before and after, every combination, line and pair, and the warnings.
- **Calibration history:** the reports found under a parent folder (by default the one above the job folder), with a plot of the angle changes over time. **View PDF**, **View ASCII**, **Explore**.
- **Export corrected LAS…** writes the used passes with the applied values, in the project's coordinate system with ellipsoidal heights, to check in other tools. Full point-cloud products come from CloudPro.

What AlignPro publishes

Every Apply writes `02_AlignPro/out/alignment.json` for CloudPro, with `alignment_report.txt` and `alignment_combinations.csv` beside it. CloudPro reads only these files.

Field	Content
mount_deg	applied roll, pitch, heading with SDEV; source calibration or manual adjustment
mount_recorded_deg	the mounting recorded in the capture

Field	Content
lidar_lever_m	lever arm used, its source (capture, measured, solved) and the recorded one
latency_s	Lidar time offset applied
boresight_deg	correction relative to the recorded mounting, with SDEV
quality	mean error now and with the recorded angles, SDEV, lines and patches used, warnings
trajectory	which trajectory it was solved on, with PosPro's id

Out of date. AlignPro's pill in the chain bar turns amber (⚠ stale) when:

- PosPro publishes a new trajectory;
- a raw capture file changes;
- the job's starting Lidar mounting is edited.

CloudPro turns stale when AlignPro publishes again. A stale result is still usable; calibrate and apply again when it matters.

One at a time. While AlignPro calibrates or publishes it holds the stage lock, so a second QGIS cannot run AlignPro on the same project. A failed run keeps the last published result.

Conventions

AlignPro uses the mapper's own conventions, so its numbers go into the mapper unchanged.

- **Vehicle frame:** X forward, Y starboard, Z down, from the Certus reference point (the IMU centre). Lever arms in metres, Forward / Starboard / Down.
- **Angles:** roll + starboard down, pitch + bow up, heading + clockwise, in degrees.
- **Lidar mounting:** M = Rz(heading) · Ry(pitch) · Rx(−roll) · Z0. At 0, 0, 0 the Lidar's rotation axis points forward and its cable connector down. Common mountings: tilted 45° = 90, 45, 0; upright with 0° forward = 0, 90, 180.
- **Near pitch 90°** (the upright mounting) roll and heading trade against each other: a tiny correction can turn 0 / 90 / 180 into numbers like −54 / 89.6 / 234. Those three together are still exact; enter all three. AlignPro warns and also gives the correction

about the vehicle axes, which stays readable.

- **Errors:** mean point-to-plane distance over the patches, in centimetres.
- **Time:** sensor time (UTC) from the captures; GPS week and leap seconds from PosPro's trajectory.

Troubleshooting and known limits

You see	Do this
AlignPro needs the DCS Core plugin	install DCS Core, restart QGIS
PosPro has not published a trajectory	process the job in PosPro; AlignPro uses the real-time trajectory meanwhile
PosPro's trajectory is out of date	re-run PosPro, then Auto calibrate and Apply
Raw data not found	project box › Relocate the job folder
Not enough planar surfaces	check overlap and range gate; widen or clear the area of interest
Lidar lever arm 0, 0, 0	enter the measured lever arm (Installation, and in the mapper)
PPS lock lost / lost packets on a line	untick that line on the Data page
Few vertical surfaces	drive past more walls, or across-road walls, for heading
Solved heading close to minus the entered one	check the heading sign before applying (below)
Wait for the current job to finish	let it end or press STOP

Known limits

- Tested on simulated mapper captures and a simulated PosPro trajectory; first real calibration drives still to come.
- The heading sign of the mounting convention is not yet confirmed on a real mounting with a non-zero heading.
- The lever arm is measured, not solved. A lever-arm solve exists in Settings for diagnosis only.

- One Lidar per project; the mapper's Hesai XT-32 format.